

VU MECHANICS – LAWI 100515 (PI)

SS 2026

2nd Exercise Exam

26.06.2026 GROUP A

Name:	
Student ID number:	
Study program code:	
Notes	
Result	____/100

Tasks	Possible Points	Achieved Points
1) Task 1 – Section Forces and Moments	40	
2) Task 1 – Truss System	30	
3) Task 2	30	

Working Time: 90 min **Passing Grade:** $\geq 50\%$

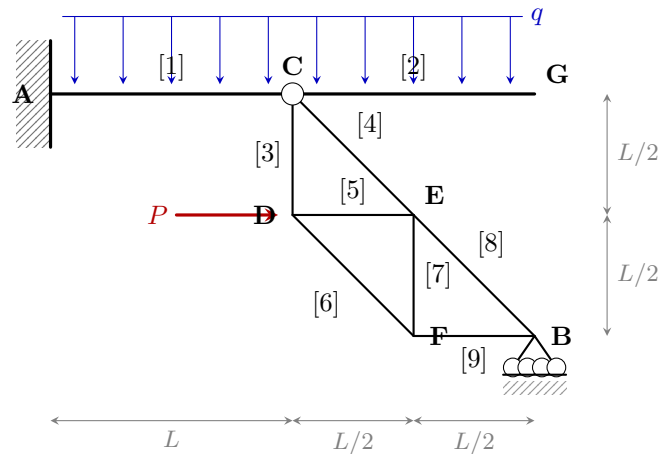
Permitted Aids: Formula sheet (boku-learn), calculator, set square, compass. No pencil, no red pen, no own paper.

1. Task

(70 %)

Given is the planar static system shown below. Dimensions and system parameters are given in the sketch. The structure is loaded by a **uniformly distributed load** q acting on beams [1] and [2], and a **concentrated load** P at truss node D.

$L = 1 \text{ m}$	$q = 6 \text{ kN/m}$	$P = 8 \text{ kN}$
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Tasks:

1. Check whether the system is statically determinate.
2. Determine the support reactions **in general form as functions of q , P , and L** . Draw the reactions into the system sketch.
3. Evaluate the support reactions for the given parameters:

$$A_V =$$

$$A_H =$$

$$M_A =$$

$$B_V =$$

4. Determine the section forces $M(x)$, $V(x)$, and $N(x)$ in beams [1] and [2] **in general form as functions of q , P , and L** . Define the range of the running variable x . Evaluate the functions at the beam start and end:

Beam start			Beam end		
M [kNm]	V [kN]	N [kN]	M [kNm]	V [kN]	N [kN]
Beam [1]					
Beam [2]					

5. Draw the distributions $M(x)$, $V(x)$, and $N(x)$ and mark the characteristic values.
6. Calculate the axial forces S_8 and S_9 using a **circular cut (Rundschnitt)** at node B, in general form as a function of P :

$$S_8 =$$

$$S_9 =$$

7. Calculate the axial forces S_4 , S_5 , and S_6 using **Ritter's cut (Ritterschnitt)**, in general form as a function of P :

$$S_4 =$$

$$S_5 =$$

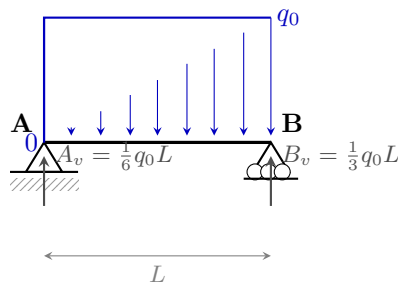
$$S_6 =$$

2. Task**(30 %)**

Given is a simply supported beam (span L) with a triangular load (0 at A, q_0 at B). The support reactions are $A_v = \frac{1}{6}q_0L$ and $B_v = \frac{1}{3}q_0L$. The bending moment distribution is:

$$M(x) = -\frac{q_0}{6L}x^3 + \frac{q_0L}{6}x \quad (0 \leq x \leq L)$$

$L = 1 \text{ m}$	$q_0 = 4 \text{ kN/m}$
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**Tasks:**

1. Determine the shear force $V(x)$ as a function of x .
2. Find the position $x_{\max}$ along the beam where the moment is maximum.
3. Calculate the corresponding maximum moment $M_{\max}$.
4. Sketch the distributions of $V(x)$ and $M(x)$ with all characteristic values.